

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Semester Examination – 2023

Course: B. Tech. Branch : Computer Science & Engineering Semester: IV

Subject Code & Name: Probability Theory and Random Processes (BTBS404)

Max Marks: 60

Date: 23-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) A card is drawn from a well shuffled pack of playing cards. Find the probability that it is either a diamond or a king.	Understand	
B) In a random arrangement of the letters of the word "MATHEMATICS". Find the Probability that all the vowels come together	Understand	
C) Derive equation of Bayes' Theorem	Application	
Q.2 Solve Any Two of the following.		12
A) Ten unbiased coin are tossed simultaneously. Find the probability of obtaining, i) exactly 6 heads ii) No head	Application	
B) A continuous random variable has probability density function $f(x) = 6(x - x^2)$ where $0 \leq x \leq 1$. Find mean, variance, median.	Evaluation	
C) A die is tossed twice. Getting 'an odd number' is termed as success. Find the probability distribution of the number of successes.	Understand	
Q. 3 Solve Any Two of the following.		12
A) Calculate probable error. If the coefficient of correlation is 0.92 and number of pairs of items are 25.	Application	
B) The ranks of some 16 students in Mathematics and Physics are as follows. Two numbers within brackets denote the ranks of the students in Mathematics and Physics: (1,1), (2,10), (3,3), (4,4), (5,5), (6,7), (7,2), (8,6), (9,8), (10,11), (11,15), (12,9), (13,14), (14,12), (15,16), (16,13). Calculate the rank correlation coefficient for the proficiencies of this group in Mathematics & Physics.	Application	
C) What is Correlation? Explain its types and causation.	Understand	

Q.4 Solve Any Two of the following.**12**

- A) Obtain the angle between the two lines of regression.
- B) From the following data of the age of husband and age of wife, find two regression lines and calculate the husband's age when wife's age is 16. Analysis

Husband age	36	23	27	28	28	29	30	31	33	35
Wife age	29	18	20	22	27	21	29	27	29	28

- C) If $\bar{x} = 8.2$; $\bar{y} = 12.4$; $\sigma_x = 6.2$; $\sigma_y = 20$; $r(x,y) = 0.9$, find the lines of regression. Estimate the value of x for $y = 10$ and estimate y for $x = 10$. Application

Q. 5 Solve Any Two of the following.**12**

- A) A random sample of size 36 is taken from a normal population with known variance $\sigma^2 = 25$. If the mean of the samples is $\bar{x} = 42.6$ test the null hypothesis $\mu = 45$ against the alternative hypothesis $\mu < 45$ with $\alpha = 0.05$ Application
- B) Explain Null Hypothesis and Alternative Hypothesis. Evaluation
- C) In a random sample of 340 students, 178 of the 210 females and 90 of the 130 males passed Statistics and Probability on their first take. Construct a 90% confidence interval for the population proportion of students who passed the subject. Application

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